# Discrete Structures

Lecture 1

# Theorems and proofs

## What are the common/significant elements of the proofs?

- Rules of inference: Logic e.g.,
  - if $p$ is true, and $p$ implies $q$, then $q$ is true.
  - if $p$ is true, then $p \vee q$ is true.
  - if $p$ is true and $q$ is true, then $p \wedge q$ is true.
  - if $p$ implies $q$ and $q$ implies $r$, then $p$ implies $r$.
  - if $p \vee q$ is true and $p$ is false, then $q$ is true.
- Strategies: Multiple strategies/techniques such as Direct, Proof by Contrapositive or Contradiction, Vacuous, Trivial, Proof by Cases, Existence and Uniqueness Proofs, Mathematical Induction, Set Equivalence
- Axioms: A statement accepted as true without proof, often serving as a starting point for reasoning and building theorems e.g., Peano's axioms, Euclid's axioms

# Proof Strategies/Techniques

1. **Direct Proof**

**Structure**: Assume p is true, use logic to prove q.
**Form**: $p \rightarrow q$
**Example**:
**Theorem**: If n is an odd integer, then $n^2$ is odd.
**Proof:** Let $n=2k+1$. Then $n^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$ which is odd.

2. **Proof by Contrapositive**

Prove $p \rightarrow q$ by proving $\neg q \rightarrow \neg p$
**Note**: Sometimes easier because negations simplify expressions.
**Example:**
*If $n^2$ is odd, then n is odd.*
Contrapositive: If n is even, then $n^2$ is even

# Proof Strategies/Techniques

**3. Proof by Contradiction**

Assume the statement you want to prove is false and show this leads to a logical contradiction.

**Example**: Prove \sqrt{2} is irrational — classic contradiction proof.

If \sqrt{2} is rational there exists p/q which is fraction in lowest terms. Square which make p^2 = 2q^2 => p is even, so p is 2k. 4k^2 = 2q^2 => q is even. So p/q is not in lowest terms which contradicts.

**4. Vacuous Proof**

When p is false, p→q is automatically true.

If the hypothesis is always false, then the implication is automatically true

Key idea: For a vacuous proof, you're always in p = false, so p→q is automatically true — no actual checking of q is needed.

| A (If...) | B (Then...) | A → B | Why |
|---|---|---|---|
| T | T | T | Promise kept |
| T | F | F | Promise broken |
| F | T | T | No promise made; no rule broken |
| F | F | T | No promise made; no rule broken |

# Proof Strategies/Techniques

**5. Trivial Proof**

When q is true, p→q holds regardless of p.

| A (If...) | B (Then...) | A → B | Why |
|---|---|---|---|
| T | T | T | Promise kept |
| T | F | F | Promise broken |
| F | T | T | No promise made; no rule broken |
| F | F | T | No promise made; no rule broken |

**6. Proof by Cases**

Split the proof into separate cases (e.g., “even or odd”) and prove each individually. Exhaustive proof is a special form where each case may involve checking a finite list of possibilities e.g., Truth table.

**Example:** If n is an integer, then $n^2 \geq 0$.

Cases: $n>0$, $n=0$, $n<0$.

# Proof Strategies/Techniques

**7. Existence Proofs**

**Constructive**: Find an example explicitly.

**Nonconstructive**: Show existence without giving example (often contradiction or theorem).

**Example:** There exists a positive integer n such that $n$^2−5$n$+6=0.

**Constructive:** $n$^2−5$n$+6=0 => (n-2)(n-3) = 0 => n =2 or n = 3.

**Non-constructive:** Per Vieta's formula sum of roots (-b/a) is 5 while product is 6 (c/a) i.e., both positive and integers with a being integer, which implies a positive integer solution must exist.

**8. Uniqueness Proofs**

**Existence:** Show at least one object satisfies the property.

**Uniqueness:** Assume two objects satisfy the property, prove they must be the same.

**Example:** There is a unique real number $x$ such that 3$x$+5=0.

Existence: x = -5/3 => atleast one real x exists.

Uniqueness: 3x_1+5=0 and 3x_2+5=0. Subtract 3(x_1-x_2) = 0 => x_1 = x_2.

# Proof Strategies/Techniques

**9. Proof by Mathematical Induction**
**Ordinary Induction**: Prove for base case (say k) and an inductive step (k+1).
**Strong Induction**: Assume true for all up to ki (k0 till ki), prove for k(i+1).
**Well-ordering principle**: Every nonempty set of nonnegative integers has a least element (specifically for integers).

**10. Counterexample (Disproof)**
**When to use:** To disprove a universal claim (∀x, P(x)).
**Method:** Find one specific example where P(x) fails.
**Example:** "All prime numbers are odd." Counterexample: 2

**11. Set Equivalence Techniques**
When proving two sets are equal ($E1$=$E2$): Proving mutual inclusion ($E1 \subseteq E2$ and vice versa), Using set-builder notation and logical equivalences, or Employing membership tables.

# Example Theorems

1. $\neg(p \wedge q)$ is logically equivalent to $\neg p \vee \neg q$
2. For all $a, b, c \in \mathbb{R}^{\geq 0}$, if $a^2 + b^2 = c^2$, then $a + b \geq c$.
3. If 6 is prime, then $6^2 = 30$.
4. $x$ is an even integer iff $x + x^2 - x^3$ is even.
5. There are infinitely many prime numbers.
6. There exist irrational numbers $x, y$ such that $x^y$ is rational.
7. For all $n \in \mathbb{N}$, $n! \leq n^n$.
8. There does not exist a (input-free) program which will always determine whether an arbitrary (input-free) program will halt.

# Proof for Theorem 1

Theorem 1: $\neg(p \wedge q)$ is logically equivalent to $\neg p \vee \neg q$

**Proof using Truth Table i.e., Proof by cases** (Exhaustive proof):

| $p$ | $q$ | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|---|---|---|---|---|---|---|
| T | T | T | F | F | F | F |
| T | F | F | T | F | T | T |
| F | T | F | T | T | F | T |
| F | F | F | T | T | T | T |

✅ The columns for $\neg(p \wedge q)$ and $\neg p \vee \neg q$ match exactly, so:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

# Proof for Theorem 2

Theorem 2: For all $a, b, c \in \mathrm{R}^{\geq 0}$, if $a^2 + b^2 = c^2$, then $a + b \geq c$.

**Direct proof** as follows:

(a+b)^2 = a^2 + 2ab + b^2 = c^2 + 2ab >= c^2 since a,b are positive.

# Proof for Theorem 3

Theorem 3: If 6 is prime, then $6^2 = 30$.

**Vacuous proof** as follows:

p: 6 Is prime

q: $6^2 = 30$

We are looking to prove $p \rightarrow q$

Since p is false, $p \rightarrow q$ is (vacuously) true independent of q.

# Proof of Theorem 4

Theorem 4: For all $x \in \mathbb{Z}$, $x$ is even iff $x + x^2 - x^3$ is even.

- To show "A iff B", you must show two things:

*i.* $A$ implies $B$ and

ii. its converse, $B$ implies $A$ (OR $\neg A$ *implies* $\neg B$).

**Proof in Two directions**:

- Forward direction
  - Let $x \in \mathbb{Z}$ and $x$ even.
  - i.e., $x = 2k$ for some $k \in \mathbb{Z}$.
  - Then $x + x^2 - x^3 = 2k + 4k^2 - 8k^3 = 2(k + 2k^2 - 4k^3)$ which is even.
- Reverse direction
  - We will show contrapositive i.e., $x$ is not even $\Rightarrow$
    - $x + x^2 - x^3$ is not even, i.e., $x$ is odd $\Rightarrow$ $x + x^2 - x^3$ is odd.
  - Let $x \in \mathbb{Z}$ be odd, i.e., $x = 2k + 1$ for some $k \in \mathbb{Z}$.
  - Then $x + x^2 - x^3$ is odd (expand with 2k+1). Hence proved.

# Proof for Theorem 5

Theorem 5: There are infinitely many primes.

**Proof by contradiction:**

1. Suppose there are only finitely many primes, say $p_1 < p_2 < \ldots < p_r$.
2. Let $k = (p_1 * p_2 * \ldots * p_r) + 1$.
3. When $k$ is divided by any $p_i$, it has remainder 1. So none of the $p_i$'s divide $k$.
4. But $k > 1$ and $k$ is not prime (since $k > p_r$), so $k$ can be written as a product of primes
   - Fundamental theorem of arithmetic: Any natural number $> 1$ can be written as a (unique) product of primes.
5. Now take any prime $p$ in this product, then, $p$ divides $k$. So, by 3. above, $p \notin \{p_1, \ldots, p_r\}$.
6. This contradicts 1, since we had assumed that $\{p_1, \ldots p_r\}$ was the set of all primes. $\square$

# Proof for Theorem 6

Theorem 6: There exist irrational numbers $x$ and $y$ such that $x^y$ is rational.

**Non-constructive Proof:**

- Consider $\sqrt{2}$. First show that $\sqrt{2}$ is irrational.
- Let x = y = $\sqrt{2}$ and consider z = $\sqrt{2}^{\sqrt{2}}$.
- Case 1: If z is rational, we are done.
- Case 2: Else z is irrational.
  - Then consider $z^{\sqrt{2}}$ = $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}}$ = $(\sqrt{2})^2$ = 2
  - Thus, we have found two irrationals x = $\sqrt{2}^{\sqrt{2}}$, y = $\sqrt{2}$ such that $x^y = 2$ is rational.

Note that the proof is not constructive since we do not determine whether $\sqrt{2}^{\sqrt{2}}$ is irrational – can we do a Constructive proof ?

# Axioms

A statement accepted as true without proof, often serving as a starting point for reasoning and building theorems e.g.,

- Euclid's axioms for geometry in 300 BCE e.g., from any point to any other point, a straight line can be drawn [drawing a straight-line axiom].
- Peano's axioms for natural numbers in 1889 e.g., every natural number has a successor.
- Zermelo-Fraenkel and Choice axioms (ZFC) are a small set of axioms from which a lot of mathematics can be inferred and are widely used
  - They describe how sets behave, what kinds of sets exist etc.
  - But proving say $2+2=4$ requires Thousands of lines of proof.

# Mathematical Induction

## Induction (Axiom)

Let *P(n)* be a property of non-negative integers. If

- *P(n0)* is true (Base case)
- For all $k \geq n0$, $P(k) \Rightarrow P(k+1)$ (Induction Step), then $P(n)$ is true for all $n \in \mathbb{N}$.

# Proof for Theorem 7

Theorem 7: For all integers $n > 1$, $n! < n^n$

Proof by induction: we will show for all $n \geq 2$, $n! < n^n$

1. Base case For $n = 2$, $2! = 2 \leq 4 = 2^2$, so Base Case is true.
2. Induction Hypothesis: Assume, for some $n = k \geq 2$, $k! < k^k$
3. Induction step: To show: $(k + 1)! \leq (k + 1)^{(k+1)}$

$$(k + 1)! = k! \cdot (k + 1) \leq k^k (k + 1) \text{ (by Induction Hypothesis)}$$
$$< (k + 1)^k \cdot (k + 1) = (k + 1)^{(k+1)}$$

4. Hence by induction, we conclude that for all $n \geq 2$, $n! < n^n$.

# Examples by induction

- Summation: For every positive integer $n$,

$$1^2 - 2^2 + 3^2 - \cdots + (-1)^{n-1} n^2 = (-1)^{n-1} \frac{n(n+1)}{2}$$

- Inequality: If $h > -1$, then $1 + nh \leq (1 + h)^n$ for all non-negative integers $n$.

# Strong Induction

Theorem: Every integer $n>1$ can be written as a product of prime numbers.

- If you try **weak induction**, you only get to assume the statement for $n=k$ when proving it for $n=k+1$.
- If $n+1$ is composite, you'll need the property for **both** of its smaller factors — which could be much smaller than $n$.
- Hence need strong induction: you assume the property for *all* numbers less than your target, so you can apply it to *any* factorization step.

# Strong Induction

Proof for "Every integer $n>1$ can be written as a product of prime numbers."

- **Base case:** $n=2$ is prime, so it's a product of primes (just itself).
- **Inductive hypothesis:** Assume that every integer $m$ with $2 \leq m \leq k$ can be written as a product of primes.
- **Inductive step:** Consider $k+1$.
- If $k+1$ is prime, we are done.
- If $k+1$ is composite, then $k+1=a \cdot b$ where $2 \leq a,b \leq k$.
- By the inductive hypothesis, both $a$ and $b$ can be written as products of primes.
- Multiplying these prime products gives a prime factorization for $k+1$.
- Thus, by strong induction, every integer $n>1$ is a product of primes.

# Strong Induction vs. WOP

## Strong induction:

- You fix a number n and assume the statement holds for all smaller positive integers.
- Then you prove it holds for n itself.
- This moves upward from the base case step-by-step.

## Well-Ordering Principle:

- You argue downward: assume there is a counterexample, take the smallest counterexample, and derive a contradiction.
- You do not explicitly “step through” all numbers in order — you jump to the “first bad case” and show it can’t exist.

# Well Ordering Principle (WOP)

**Theorem:** Every integer $n>1$ can be factored into primes.

**Proof (via WOP):**

Suppose not. Let S be the set of integers >1 that are not products of primes.

If S is nonempty, WOP says it has a smallest element m [since WOP states "Every nonempty set of natural numbers has a least element."]

- m cannot be prime, because a prime itself is a product of primes (just itself).
- So, m is composite: $m=a\cdot b$ with $1<a<m$ and $1<b<m$.
- By minimality of m, both a and b are products of primes. Multiplying them gives a prime factorization for m — contradiction.

Thus, S must be empty, so every integer $n>1$ is a product of primes.

# Proof for Theorem 8

Theorem 8: There does not exist a (input-free) program which will always determine whether an arbitrary (input-free) program will halt [Halting Problem].

More formally: There is no algorithm H that, given the description of *any* program P with no inputs, always correctly answers "halts" or "does not halt" and itself always halts.

**Proof by Contradiction:**

Suppose such a program $H(P)$ exists:

- Input: code of a program $P$ (no input to P itself)
- Output: YES if P halts, NO if P runs forever
- H itself always halts.

# Proof for Theorem 8

```
function P():
   if H(P) == YES:
      loop forever
   else:
      halt immediately
```

- P runs H on *its own code*.
- If H predicts P halts → P runs forever.
- If H predicts P runs forever → P halts.

# Proof for Theorem 8

**Contradiction**

Now ask: what does H(P) output?

- If H(P)= YES (predicts halt) → by definition, P loops forever → prediction wrong.
- If H(P)= NO (predicts infinite loop) → P halts immediately → prediction wrong.

Either way, H makes a wrong prediction for P, contradicting our assumption that H works for all programs.

Hence, no such H can exist. Proves:
"Halting Problem" is undecidable even for input-free programs.